



Technische
Universität
Braunschweig

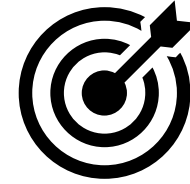
**Decision
Support**
Institut für Wirtschaftsinformatik



Data Driven Decision Making

Lecture 2 – Information and Decision Modeling

Goals of today's lecture



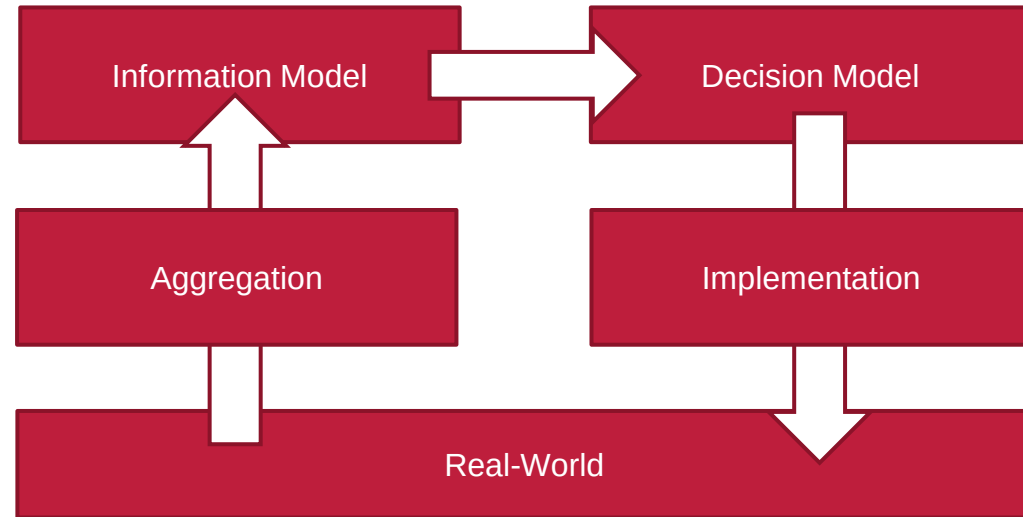
- The Process of Business Analytics
 - Information and Decision Modeling
 - Vehicle Routing with time-dependent Travel Times
 - Data comes first...
-
- Warning: In this lecture, we will use many different methods. They will not be defined in detail, but only their idea.

Agenda

- Information Modeling and Decision Modeling
- Case Study: Traveling salesperson with time-dependent travel times

Decision making

- **Real-World:** Actions are performed and resulting states can be observed
- **Aggregation:** Transfers real-world data into dimensions of information model
- **Information Model:** Depicts information in a condensed way
- **Decision Model:** Defines decision space and evaluates decision
- **Implementation:** Translates solution of the decision model into real-world actions

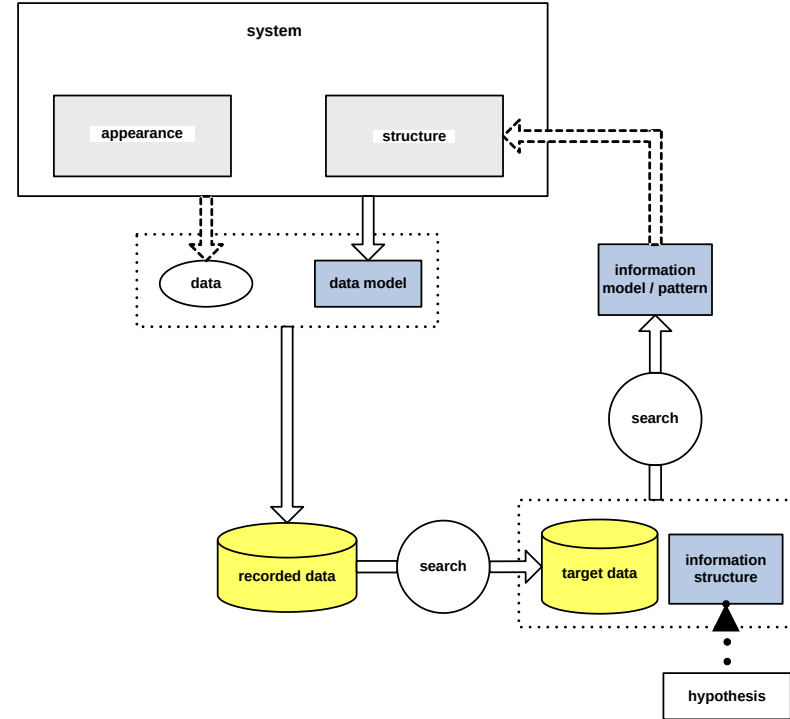


Selection of information and decision model

- Information and decision model are designed based on initial hypotheses
- Hypotheses base on observed behavior of the real-world problem and data (descriptive analytics)
- The hypotheses are not necessarily correct.
- The quality of the hypotheses can be evaluated by implementing the solution of the decision model
- Observations may lead to an update of the hypotheses, and consequently, information and decision model

Intelligent Data Analysis: From Appearance to Structure

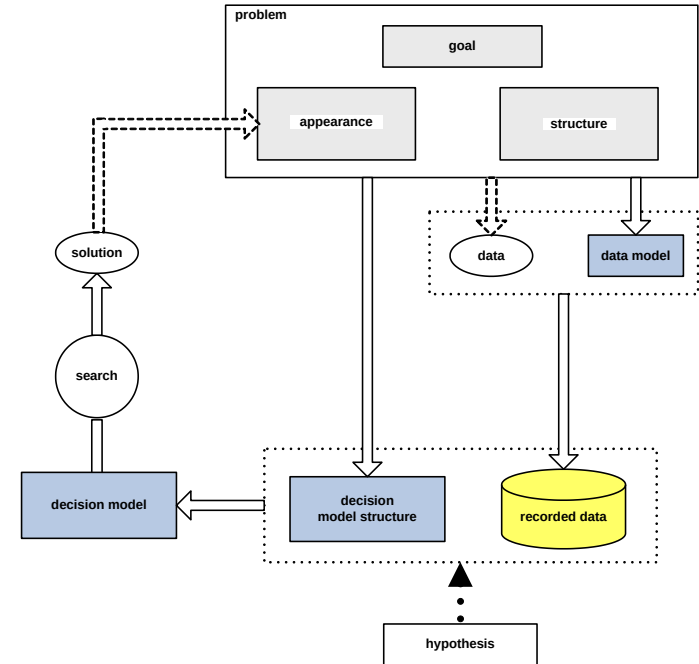
- Data from any real-world system just reflects its appearance. The structural properties are hardly visible (Google Maps shows congestion, not reason)
- Appearance may be incomplete, faulty, exceptional
- System structure helps building the data model (do margarine sales depend on visits or vice versa?)
- Recorded data are just observations of general system appearance (data $\approx$ tiny no. of samples)
- Recorded data is to be cleaned, corrected, imputed, validated and thinned into a target data set
- Data Mining computes model from target data set
- A DM concept implements our hypothesis (assumption) about statistical correlation in the data
- Model/pattern $\rightarrow$ evidence on system structure



Meisel, Mattfeld (2010): Synergies of operations research and data mining, EJOR vol 206, pages 1-10

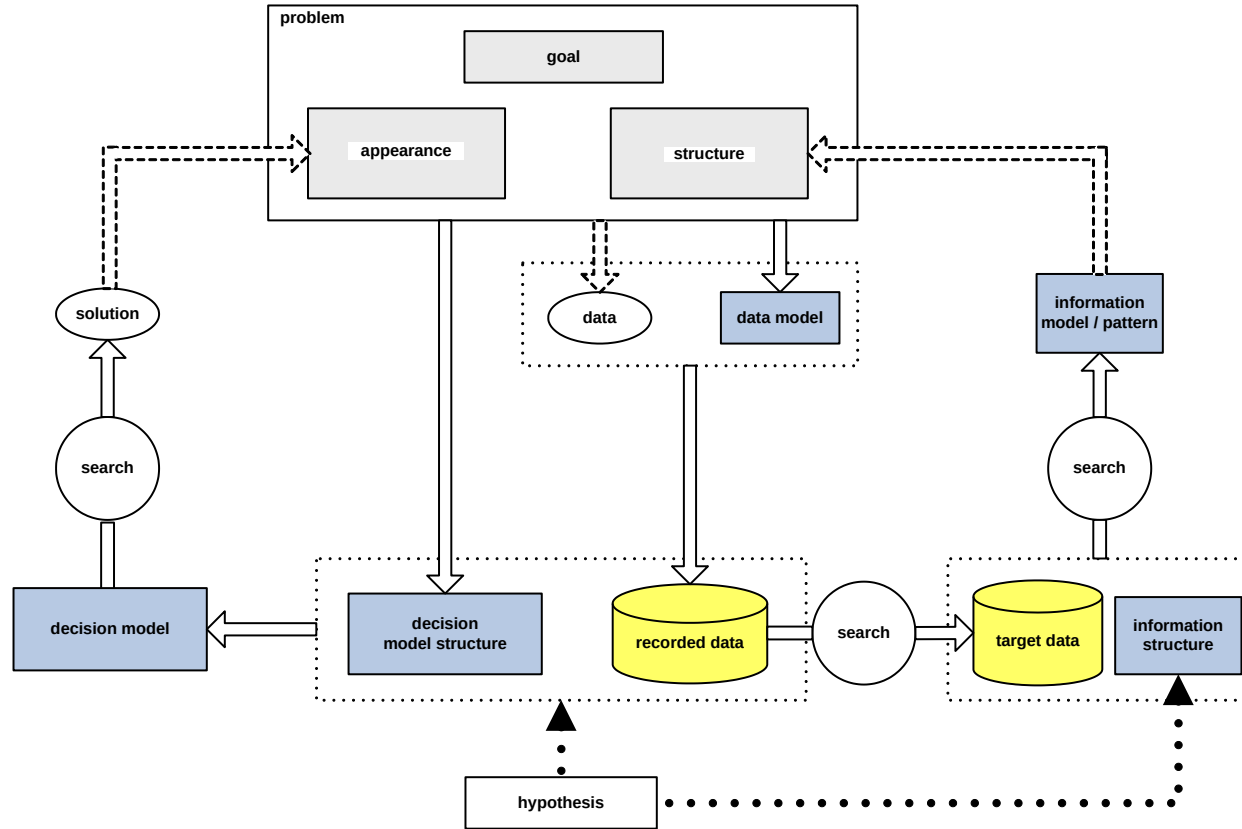
Planning of Mobility/Transp.: From Structure to Appearance

- The real-world decision problem has to be simplified in order to meet the requirements of math. modeling
- Therefore important features of the problem structure are selected by means of a hypothesis
- Features are only considered if data is available
- Important features are objective, decision space, constraints
- The general structure of the decision model helps to bridge the path from problem → decision model
- Decision model structures are e.g. Assignment, Routing, Scheduling
- The resulting solution depicts one (optimal) appearance of the problem → evaluation?



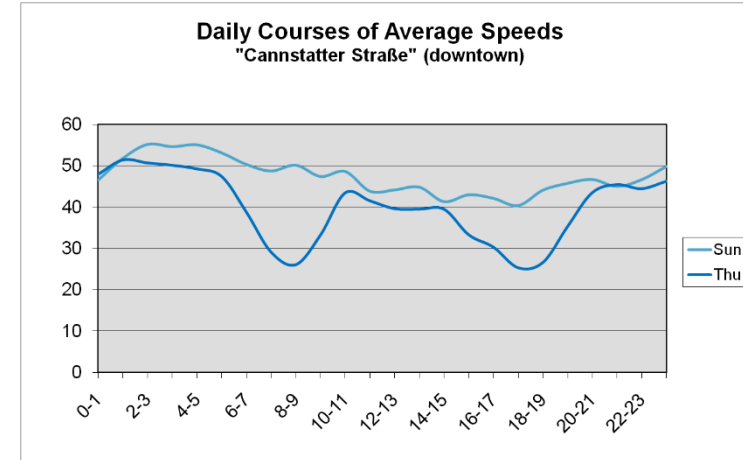
Meisel, Mattfeld (2010): Synergies of operations research and data mining, EJOR vol 206, pages 1-10

Unified View of IDA and PMT



Example: Traveling Salesperson Problem

- Hypotheses: Travel times are constant
- Information model: travel time matrix
- Decision model: mixed integer program without temporal considerations
- Implementation leads to unexpected results (usually, more time is required than planned)
- Hypotheses are adapted to capture time-dependent travel times leading to changes in
 - Information model: A set of travel time matrices (and some transition considerations)
 - Decision model: mixed integer program with time-of-day considerations (x_{ijt})

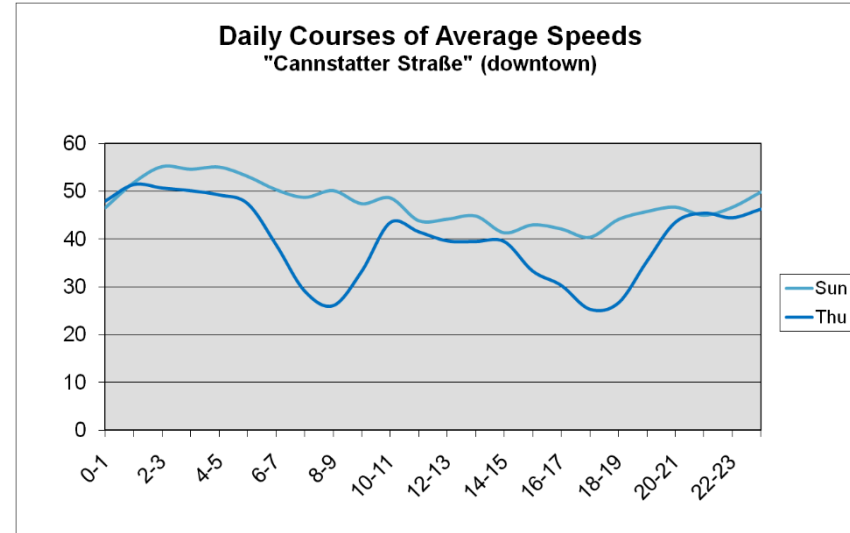


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- Case Study: Traveling salesperson with time-dependent travel times

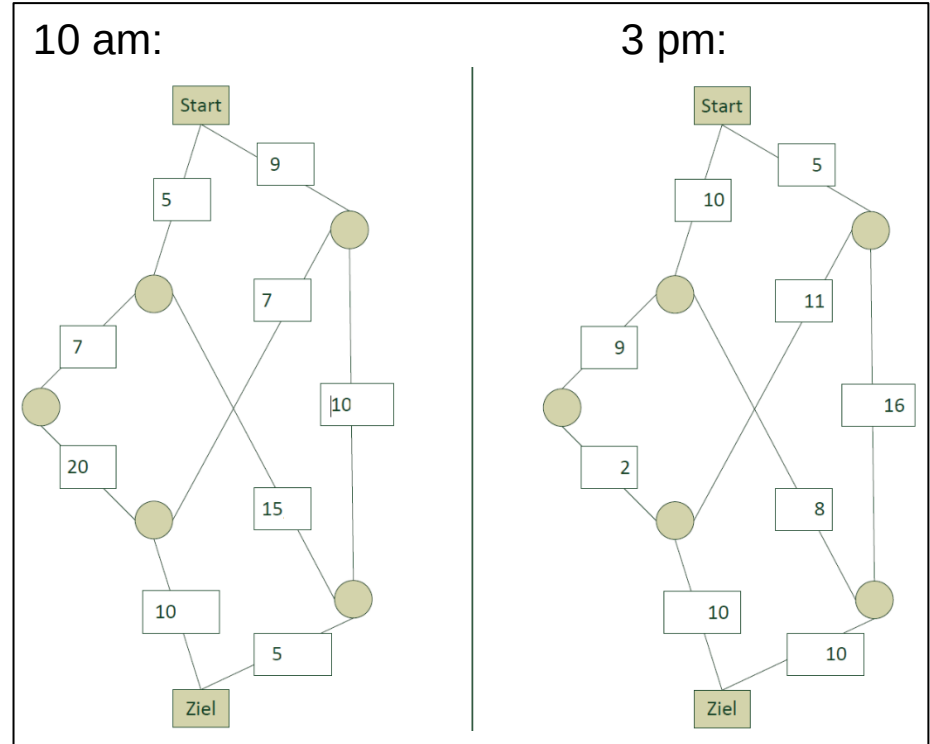
Motivation

- Service Routing
- Save costly driving time
- Temporal commitment:
 - Increase customer service experience
 - Ensure customer availability
- Both require accurate consideration of travel times
- Source: Jan Ehmke, *Integration of Information and Optimization Models for Routing in City Logistics*, Springer



Time-Dependent Travel Times

- Varying traffic over time and space
- Different travel times per street segment
- Different shortest paths between customers
- Different sequences of customers depending on start time of the tour



Modeling (sketched)

- Information Model:
 - Time-dependent travel times between customers (including depot)
 - Time-dependent travel time matrix, or a set of travel time matrices
- Decision Model:
 - Objective: Minimize travel times
 - Constraints:
 - Serve all customers
 - Routing-constraints (Subtour-elimination, etc.)
 - Decision variables: Travel between customers i and j **at time t**

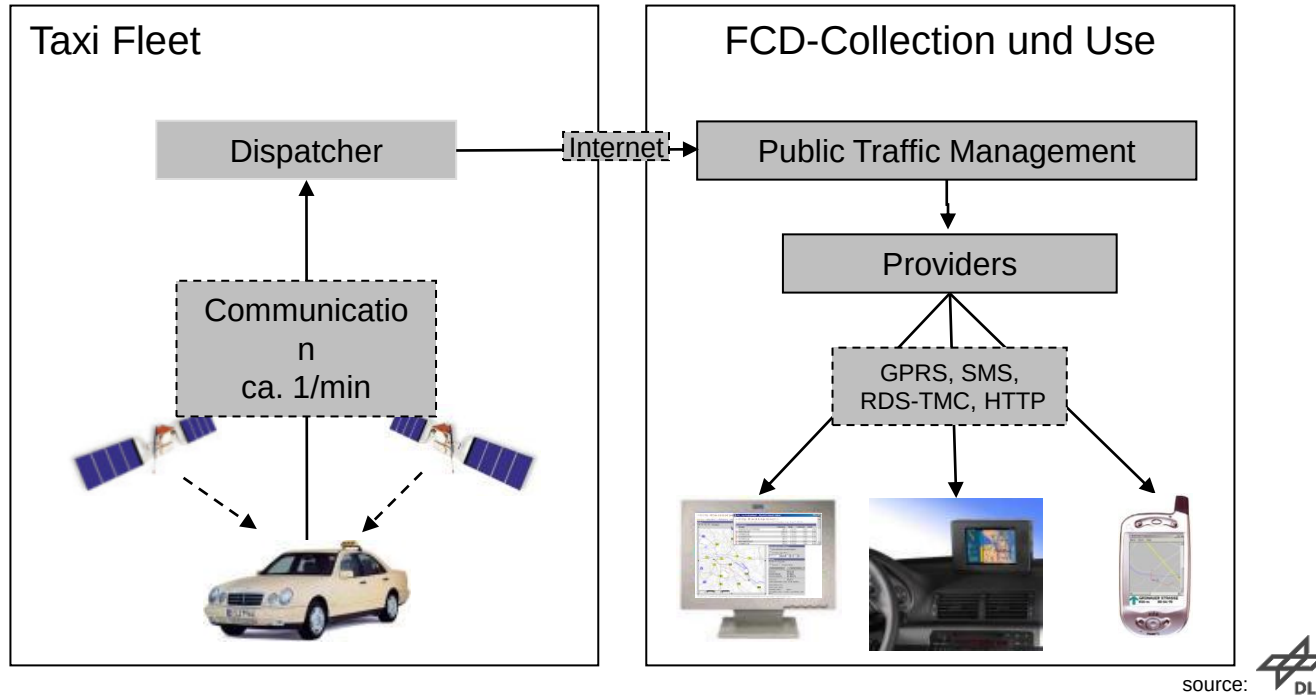
Data Collection: Floating Car Data

- Travel times per time and street segment
- Large quantities required to calculate accurate mean values
- Floating Car Data: A large fleet of vehicles sends their position, direction, and potentially their speed in frequent time intervals

Google Maps gets in a jam in central Berlin thanks to artist with 99 mobile phones towed in handcart

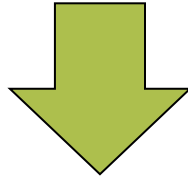


Data Collection: Floating Car Data (Public)

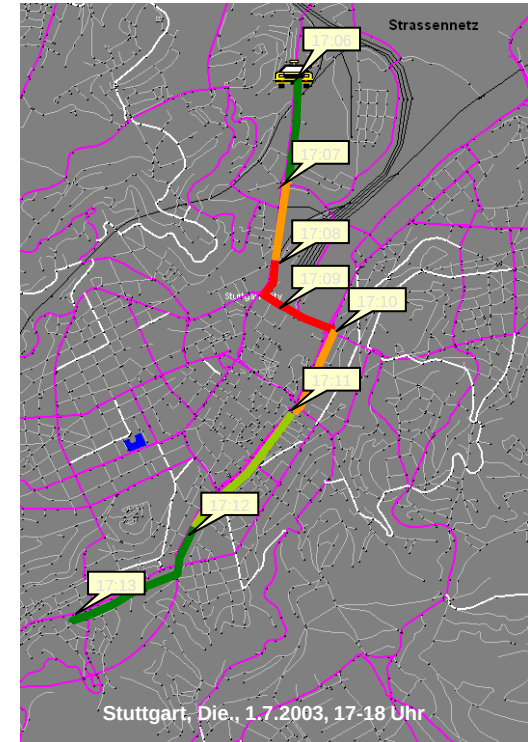


Floating Car Data: Speed Calculation

- Taxi Data does not provide street, direction, and speed
- Required steps:
 - Observe taxi positions
 - Project position to a street segment
 - Calculate observed speed for this segment



Link.ID	Zeit	v
1234	03.02.2005 12:43:30	50 km/h



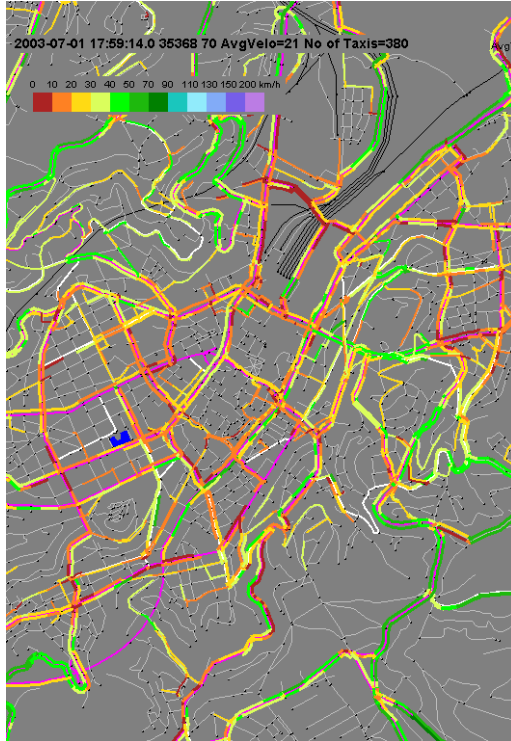
source:



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Floating Car Data: Speed Calculation



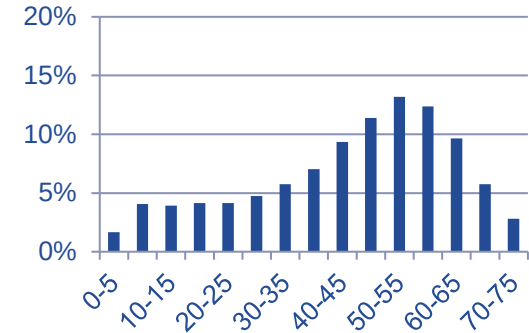
link	time	speed
-53197590	2003-03-24 00:11:19	55
-53197590	2003-03-24 00:07:35	61
-53197590	2003-03-24 00:21:25	49
-53197590	2003-03-24 00:22:49	20
-53197590	2003-03-24 01:48:16	64
-53197590	2003-03-24 01:22:19	70
-53197590	2003-03-24 02:06:48	43
-53197590	2003-03-24 02:26:25	49
-53197590	2003-03-24 02:52:32	29
-53197590	2003-03-24 03:17:53	61
-53197590	2003-03-24 03:18:23	63
-53197590	2003-03-24 03:21:59	40
-53197590	2003-03-24 04:34:33	67
-53197590	2003-03-24 04:43:56	41
-53197590	2003-03-24 05:14:18	74
-53197590	2003-03-24 04:48:38	61

source:  DLR

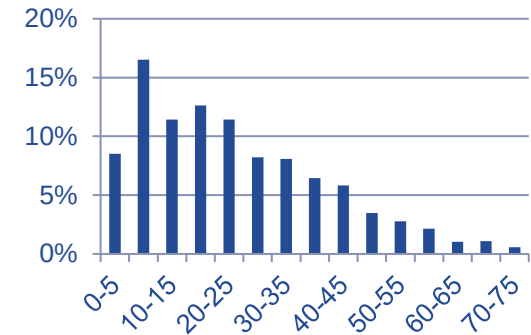
Data Analysis

- Varying speed patterns over the day (confirms hypothesis)
- Variation even at the same day, same time (ignored by our hypothesis, next lecture)
- There might be outliers or errors (taxi data!)

Downtown link, Monday



Downtown link, Mon 8-9 am



Data Cleaning

- Which observations are outliers or errors?
- Objective evaluation difficult since there are many external factors (e.g., traffic lights, weather, ...)
- Example for data cleaning:
Filter all observations with speed higher than 1.5x speed limit.

link	time	speed	maxspeed
-53197590	2003-03-24 00:11:19	55	75
-53197590	2003-03-24 00:07:35	61	75
-53197590	2003-03-24 00:21:25	49	75
-53197590	2003-03-24 00:22:49	20	75
-53197590	2003-03-24 01:48:16	64	75
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-53197590	2003-03-24 05:14:18	74	75
-53197590	2003-03-24 04:48:38	61	75

Data Visualization

- Spatial distribution of observations
- Some areas are well-covered
- Some areas only have a few observations (taxi data)
- Same is true for time of day
- Can we trust our observations?

→ Aggregation and generalization needed



Small number of observations

Large number of observations

Aggregation

1. Define time-steps W
(e.g.,: 24 x 7)

2. Calculate mean travel speed

- per segment l
- in time-step w
- from observation v_{lwi}

Link.ID	Zeit	v
1234	03.02.2005 12:43:30	50 km/h

$$v_{lw} = n / \sum_{i=1}^n \frac{1}{v_{lwi}}$$

$W = 1$: One value per segment
 $W = 24 \times 7$: Per day 24 values per segment

Zeit	0-1	1-2	...	3-4	...	8-9	...	16-17	...	22-23	23-24
So	47	52	...	55	...	50	...	42	...	47	50
Fr	47	52	...	52	...	33	...	31	...	44	46

source: DLR

Generalization

- Aggregation results in millions of values
- Some values base only on a few observations
- Decision model requires information model easy to access (and reliable)
- Data visualization reveals similar traffic quality patterns on different street segments (not shown in this lecture)
- Idea: Group street segments based on their typical traffic quality

Generalization: Normalization

- Different segments have same pattern but different absolute values (e.g., speed limit of 30 or 50)
- Normalization with maximum daily value to allow comparison

Zeit	0-1	1-2	...	3-4	...	8-9	...	16-17	...	22-23	23-24
So	47	52	...	55	...	50	...	42	...	47	50
Fr	47	52	...	52	...	33	...	31	...	44	46

Zeit	0-1	1-2	...	3-4	...	8-9	...	16-17	...	22-23	23-24
So	0.4	0.8	...	1.0	...	0.7	...	0.1	...	0.4	0.7
Fr	0.8	1.0	...	1.0	...	0.2	...	0.1	...	0.6	0.7



Generalization: Clustering

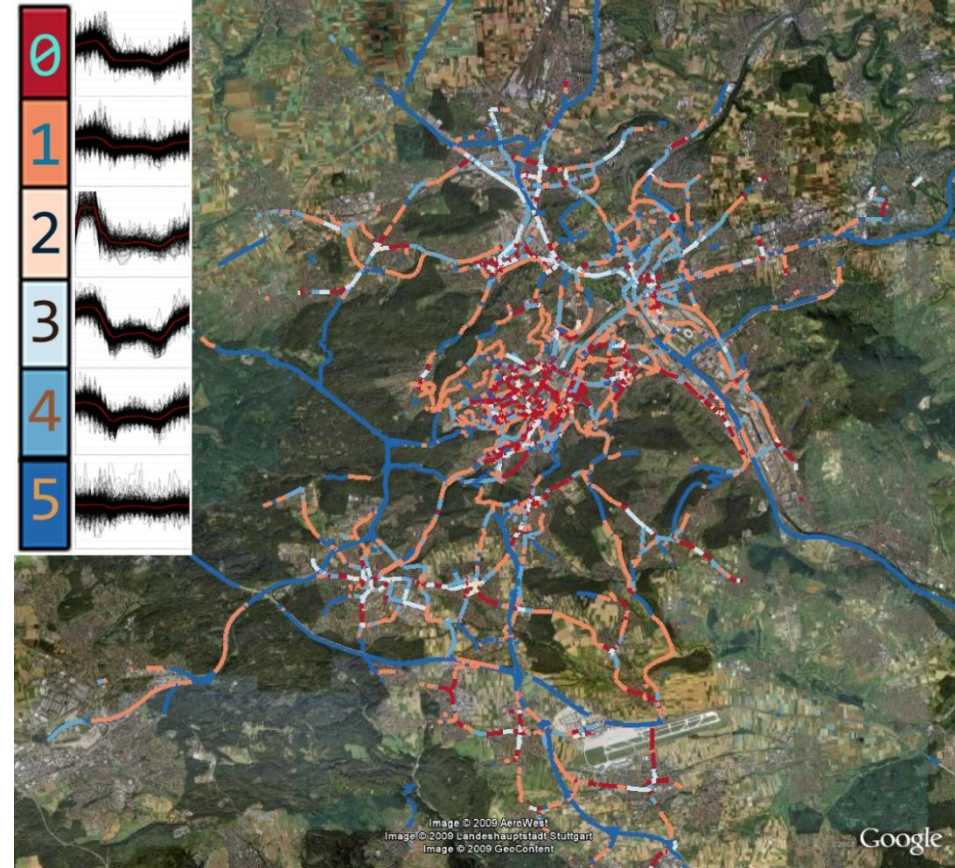
- Clustering algorithm to group similar daily developments
- 6 Clusters

Time Cluster	0-1	1-2	...	3-4	...	8-9	...	16-17	...	22-23	23-24
1	0.71	0.76	...	0.79	...	0.25	...	0.23	...	0.59	0.65
2	0.64	0.65	...	0.63	...	0.52	...	0.53	...	0.64	0.64
3	0.43	0.46	...	0.55	...	0.30	...	0.27	...	0.38	0.41
4	0.78	0.76	...	0.77	...	0.48	...	0.45	...	0.71	0.75
5	0.83	0.81	...	0.81	...	0.33	...	0.70	...	0.83	0.81

- 0 = poor traffic quality → long travel time
- 1 = good traffic quality → short travel time

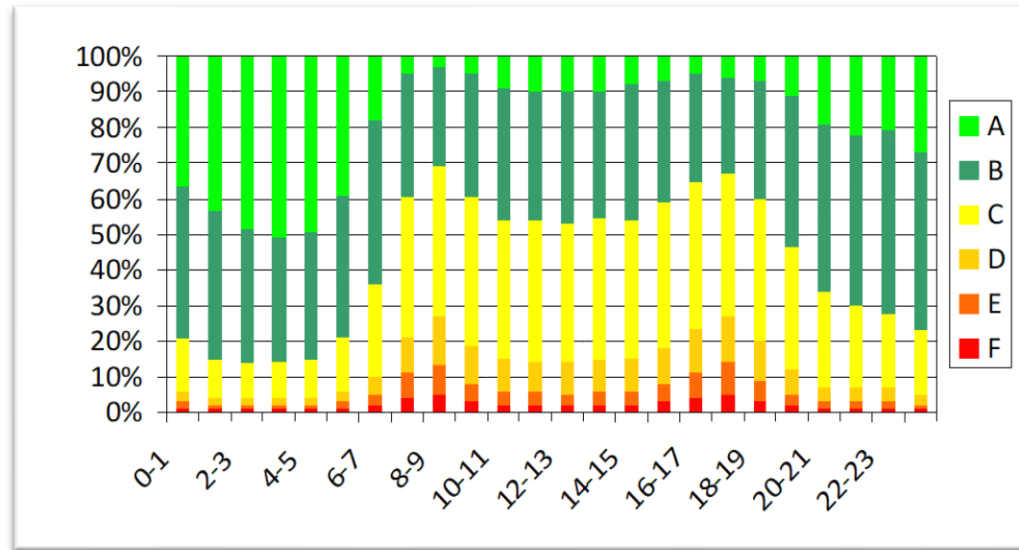
Spatial Validation

- Clusters differ in their peak behavior
- Clusters represent different street segment types, e.g.,
 - Downtown
 - Commercial
 - Highway
 - Conveyor belt



Temporal Validation

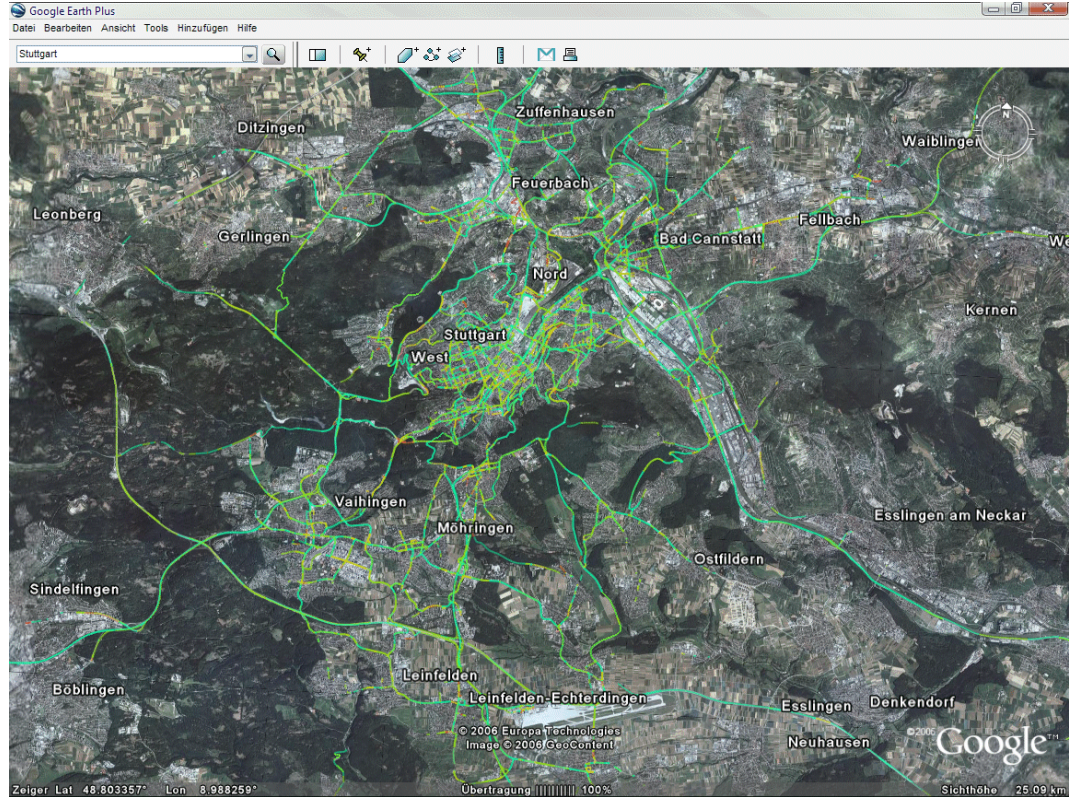
- Traffic quality over the course of the day (A – high, F – poor)



Spatial-Temporal

Monday,
3-4 a.m.,
90 % of the
segments are
free

Free
traffic

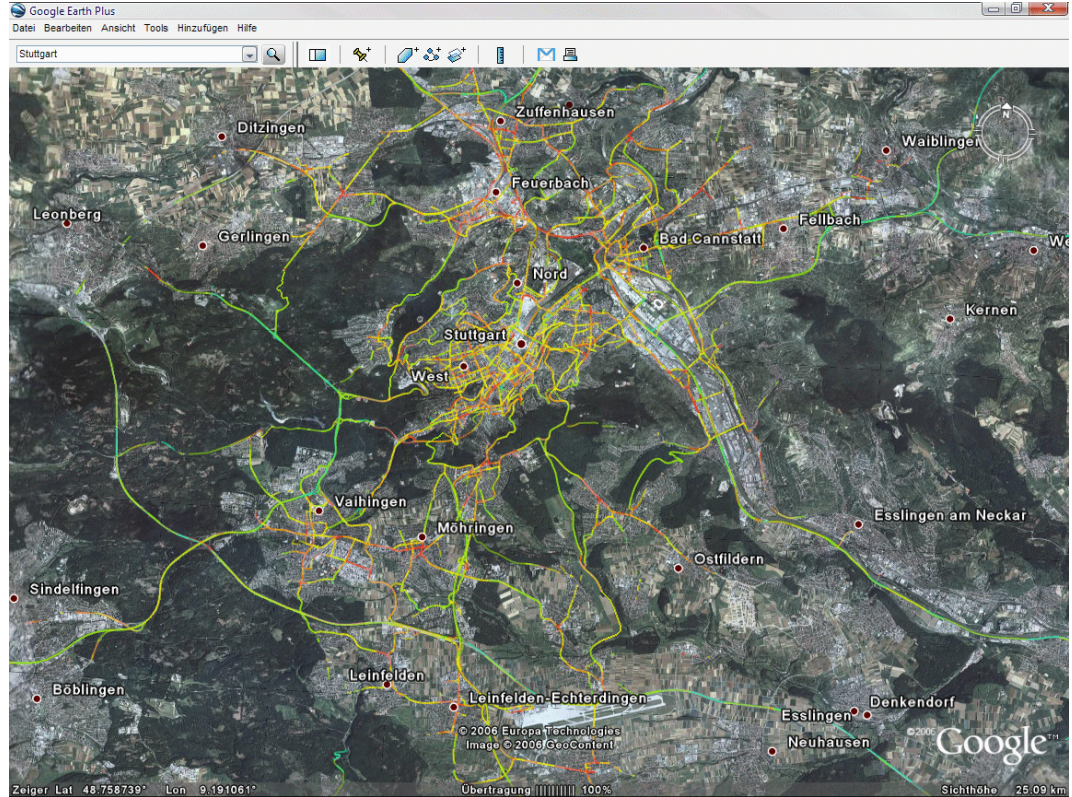


source: Google Earth

Spatial-Temporal

Monday,
5-6 p.m.,
30% are
congested

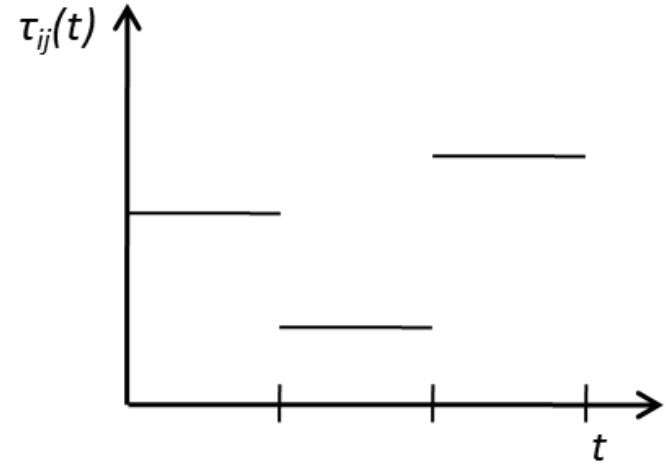
Rush hour



source: Google Earth

Information Model: Conclusion

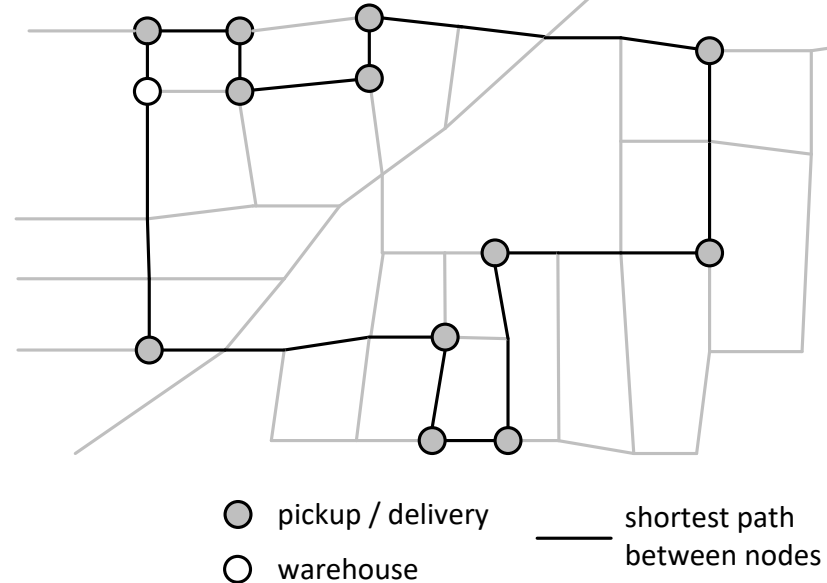
- We have a travel time matrix for each street segment and each hour
- Example: First hour longer travel, second hour shorter travel, third our longest travel time.
- How can we route based on this information?



From Segments to Paths

- There are several potential paths between customers
- Shortest paths are calculated via Dijkstra-algorithm
- We need to consider time-dependency, i.e., we are looking for time-dependent shortest paths

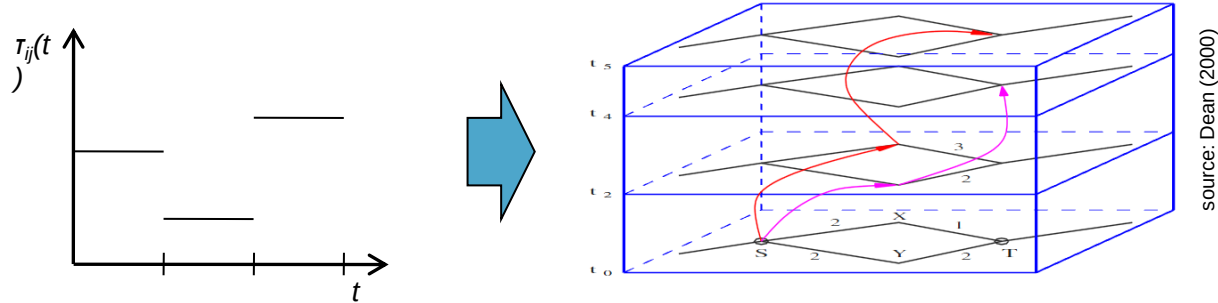
Routing



source: in Anlehnung an Rodrigues, Comtois und Slack (2009)

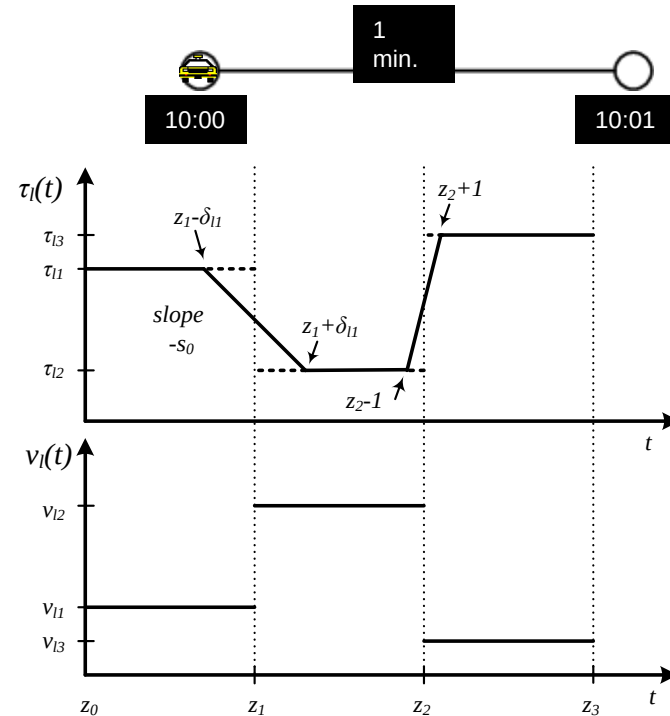
From Segments to Paths

- If we start at different times, we may take different segments
- The time-dependent Dijkstra takes this into account



From Segments to Paths

- Hard matrix breaks lead to a weird transition phenomenon
- FIFO: We need to ensure that we do not „overtake“ when starting later
- We smooth the breakpoints with some linear combinations

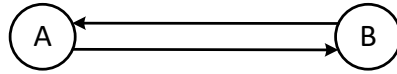


source: Fleischmann et al. (2004):
Time-Varying Travel Times in Vehicle Routing

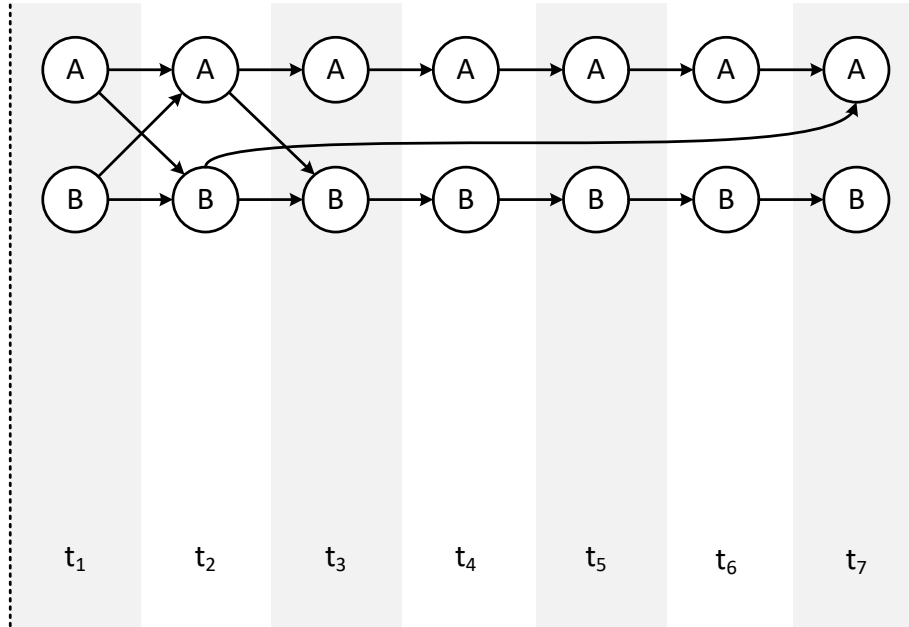
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Time-dependent Decision Variable

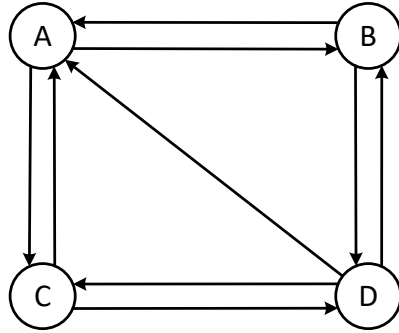


$$d_{A,B} = \{1, 1, -\} \quad d_{B,A} = \{1, 5, -\}$$



Pallotino (1997)

Time-dependent Decision Variable



$$d_{A,B} = \{1, 1, -\}$$

$$d_{A,C} = \{2, -, 2\}$$

$$d_{B,D} = \{2, 2, 2\}$$

$$d_{C,D} = \{1, -, 4\}$$

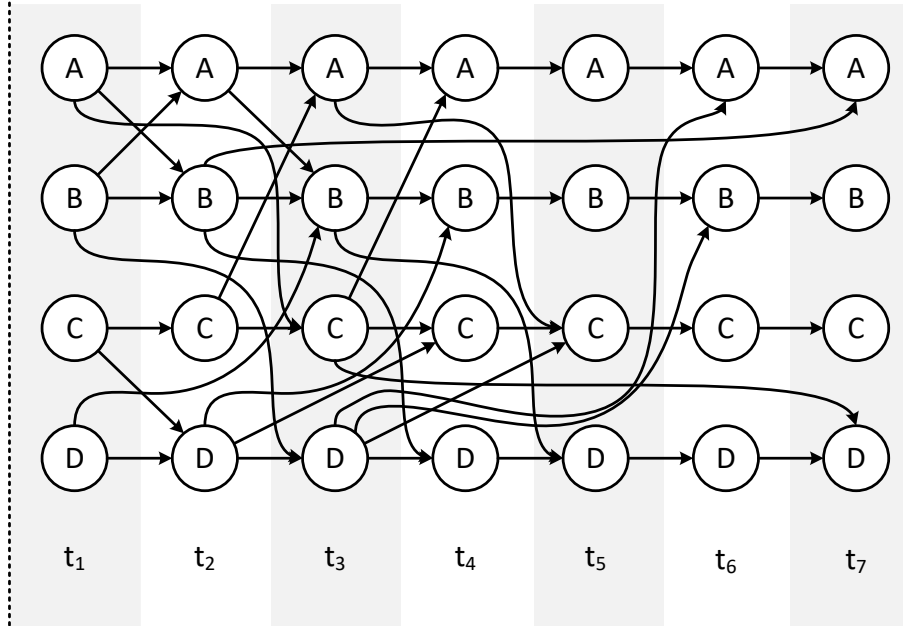
$$d_{D,A} = \{-, -, 3\}$$

$$d_{B,A} = \{1, 5, -\}$$

$$d_{C,A} = \{-, 1, 1\}$$

$$d_{D,B} = \{2, 2, 3\}$$

$$d_{D,C} = \{-, 2, 2\}$$



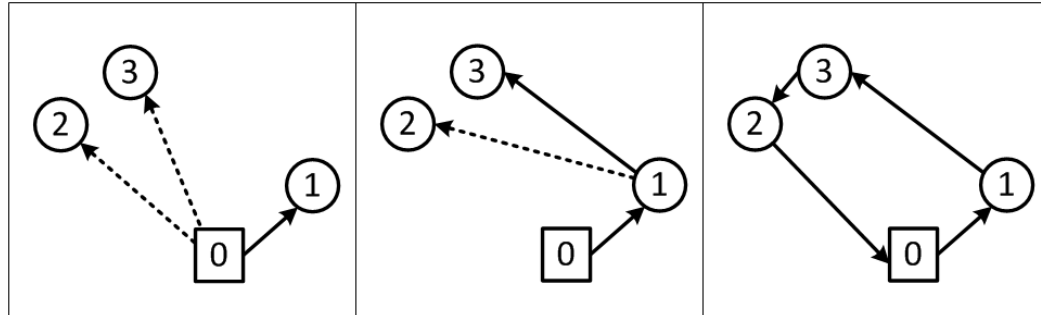
Pallotino (1997)

Finding a Solution

- Optimal calculation challenging, heuristics needed
- 4 Heuristics
 - LANTIME: Metaheuristic (see Ehmke 2012)
 - Nearest Neighbor
 - Cheapest Insertion
 - Savings
- We also solve a TSP on mean values

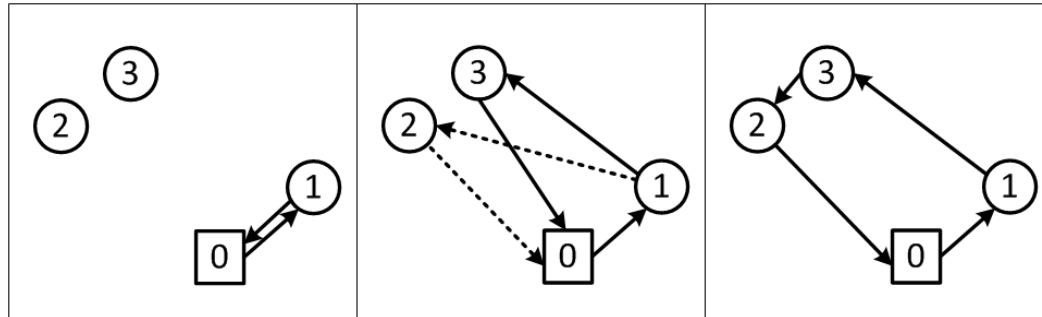
Nearest Neighbor

- Selects next customer closest to current position
- Can deal with time-dependency
- Usually leads to long travels in the end



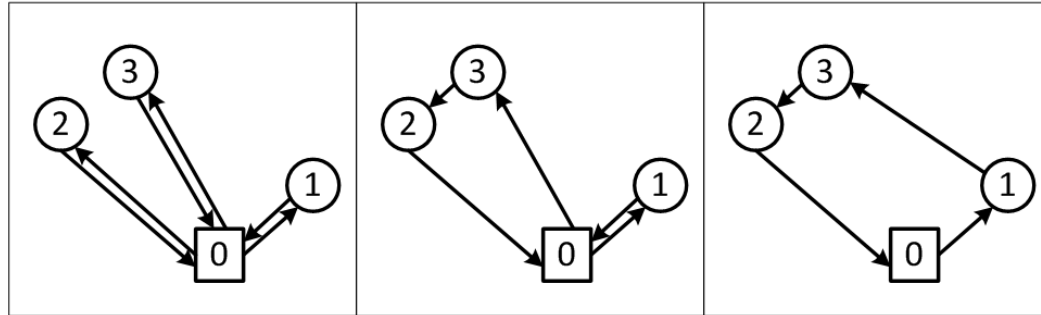
Cheapest Insertion

- Starts with empty tour
- Inserts stepwise cheapest customer in the tour
- Customers are shifted and previous decisions become ineffective due to changes in the travel times.



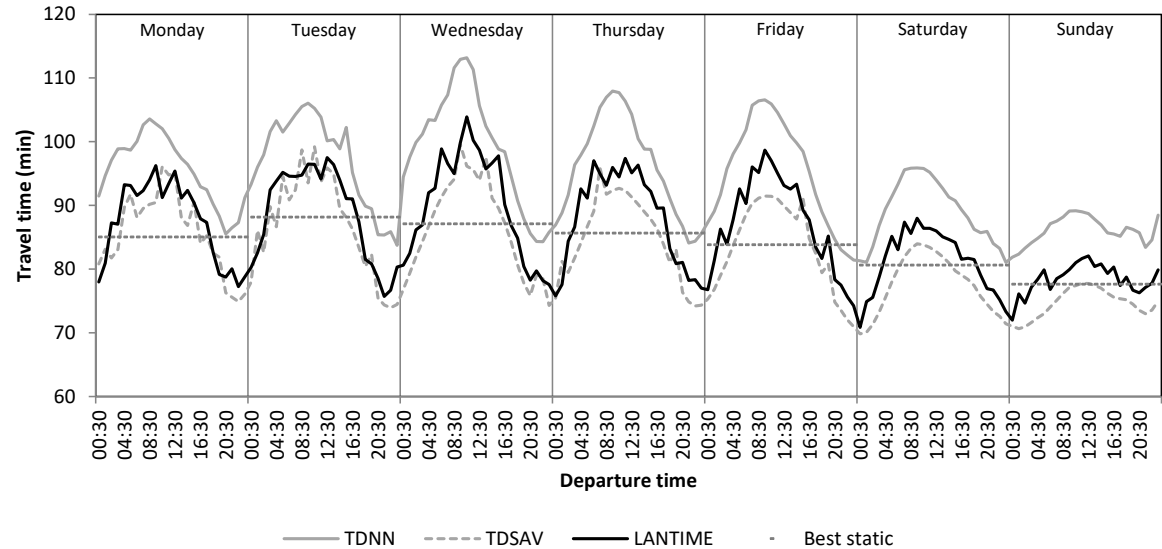
Savings

- Starts with direct-trip tours
- merges tours at minimal cost
- Time-dependency difficult to assess
- Implicitly considered at least in later steps



Results

- Different start times
- Travel time depends on time of day
- Static overestimates and underestimates



Implementation:

- Start Monday noon:
 - Avoids downtown initially
 - Serves outskirts first
- Start Monday evening:
 - Serves downtown first
 - Uses rural road in the beginning
 - Different paths



Long story short...

- Taking problem and data seriously requires a lot of work
- Information Model: Data collection, analysis, structuring, aggregation, generalization
- Decision Model: More decision variables, complex decision model, and adapted solution methodology
- Resulting approach takes more detail in modeling but assumes deterministic behavior

Outlook

Uncertainty and its impact on information and decision model

